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Late Cretaceous–Early Tertiary Tectonic History of Southern Central America

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The paleomagnetic analysis of samples from three Late Cretaceous through Eocene sites in southwesternmost Nicaragua yields pole positions close to the present pole. The results from five equivalent sites in the Pacific coastal zone of Costa Rica, on the other hand, suggest a 14° northward translation in post-Eocene time. This movement cannot have been initiated by the development of the Cocos-Nazca spreading center and it seems likely that the extensive Late Miocene plutonism in Costa Rica defines the time of collision of the Nicoya Complex (sensu largo) and the Costa Rican island arc. The Nicaraguan as well as the Costa Rican data are very different from paleomagnetic results from Honduras, and the tectonic boundary between northern and southern Central America, therefore, must lie to the north and east of the Nicaraguan sites.

INTRODUCTION

It has long been recognized that Central America consists of two distinct geological provinces [Vaughan, 1918; Schuchert, 1935; Sapper, 1937; Stille, 1940; Weyl, 1961; Dengo, 1962]. In the northern province, a Paleozoic or older metamorphic basement is overlain by mainly Mesozoic sediments. During the Tertiary, large volumes of ignimbrites were extruded. The continental character of this crust is confirmed by seismic studies [Kim, 1981]. By contrast, southern Central America seems to be built on oceanic crust. The Nicoya Complex, uplifted oceanic crust of Middle Jurassic through Cretaceous age [Schmidt-Effing, 1979], is the oldest recognized formation. It is overlain by Late Cretaceous and Tertiary open marine sediments. The major volcanic activity took place in the Late Tertiary. An excellent description of the geology of Central America is given by Weyl [1980].

The boundary between northern and southern Central America has not been established but must lie between northern Costa Rica and northern Nicaragua. Much of this area is covered by Tertiary and Quaternary volcanics and has only been incompletely mapped. In an attempt to test whether the two provinces are tectonically independent, we collected paleomagnetic samples in southern Nicaragua and in Costa Rica. Preliminary pole positions from these samples have been published as a figure by Gose et al. [1980]. By comparing these data with our results from Honduras [Gose and Swartz, 1977; Gose et al., 1980; W. A. Gose, manuscript in preparation 1983], we may be able to place constraints on the location of the boundary.

GEOLOGIC SETTING

The oldest exposed rocks in the Pacific coastal region of Nicaragua belong to the Rivas Formation, an open marine sequence of interbedded tuffaceous shale, siltstone, graywacke sandstone, and conglomerate of Cenomanian through Maastrichtian age [Tax Improvement and Natural Resources Inventory Project, 1972]. In the subsurface the Rivas Formation rests unconformably on pre-Cenomanian volcanoclastic strata. No rocks equivalent to the Nicoya Complex of Costa Rica have been recognized in Nicaragua. A depositional hiatus separates the Rivas Formation from the overlying Brito Formation. The Brito Formation is of Late Paleocene through Eocene age and is lithologically very similar to the Rivas Formation.

The Nicoya Complex is the oldest rock unit identified in all of southern Central America. In Costa Rica it is exposed along the Pacific coast, particularly well on the Nicoya peninsula. The Nicoya Complex is uplifted oceanic crust which ranges in age from Late Jurassic through Cretaceous and possibly Eocene [Schmidt-Effing, 1979; Galli-Olivier, 1979; Kuijpers, 1980]. On the Nicoya peninsula it is overlain by the hemipelagic Sabana Grande Formation of Campanian age. The Sabana Grande Formation grades into the Rivas Formation of Late Campanian through Early Paleocene age. The Rivas Formation as well as the overlying Las Palmas Formation (Lower Paleocene through Lower Eocene) consist of mainly volcanogenic turbidites and sandstones [Lundberg, 1982]. The Barra Honda Formation is a restricted unit of Paleocene reefal limestones. Most of the Eocene is represented by the Brito Formation. Detailed biostratigraphic studies by Schmidt-Effing [1979] demonstrate substantial differential vertical movement within the Nicoya peninsula.

SAMPLING AND MEASURING TECHNIQUES

In the subtropical environment of Central America, it is difficult to find suitable outcrops. Most of our sampling sites are located at places where the action of water or man provided a reasonably fresh outcrop (Figure 1). In southern Nicaragua we collected 82 samples at four sites, and in Costa Rica we drilled 324 samples at 14 sites. All but one site lie in the Pacific coastal zone because only this area has been mapped to any detail, it is accessible, and the oldest rocks seem to be restricted to this zone, at least on the surface. All samples were stored in a magnetically shielded room (100 nT) where they remained throughout the experimental procedure. A cryogenic magnetometer was used to measure the magnetic remanence. All samples were ther-

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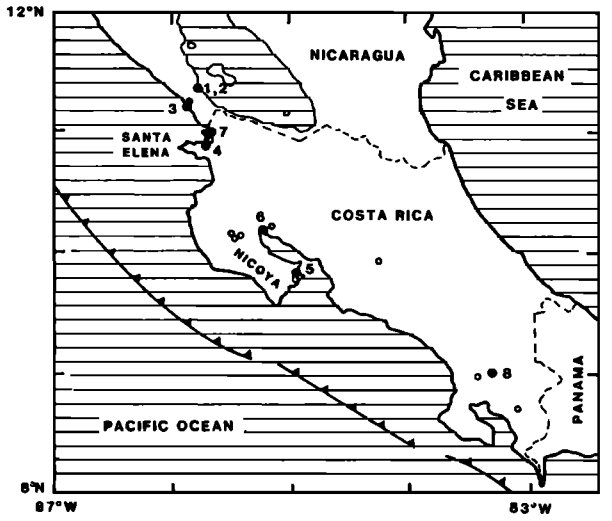


Fig. 1. Location map. Numbered solid circles are sampling sites referred to in the text and Table 1. Open circles mark sampling sites which were rejected.

mally demagnetized at least in four, usually in seven, steps. The field in the cooling region was less than 5 nT.

The directions of magnetization were analyzed using orthogonal vector projections (As-Zijderveld diagrams). After testing for the linearity of a vector trend (minimum of three sequential demagnetization steps), the mean direction is defined by the sum of vectors if the trend decays toward the origin and by the sum of subtracted vectors if it does not. A disappointingly large number of sites did not yield usable data, most likely due to the pervasive weathering. A site was rejected if either the directions of magnetization for most samples changed erratically during demagnetization or if the distribution of directions was nearly random at all demagnetization steps. On the basis of this criterion, one site in Nicaragua and nine sites in Costa Rica were eliminated.

The results from the remaining eight sites were analyzed in two steps. First, the mean direction of all samles from a given site was calculated. Directions whose angular distance from this mean exceeded 4 times the radius of the 95% circle of confidence were then eliminated and a new mean was calculated. The latter value is entered in Table 1.

RESULTS

Nicaragua (Sites 1, 2, and 3)

Along the shore of Lake Nicaragua, some 10 km southeast of the city of Rivas, we sampled the Rivas Formation at two sites about 3 km apart. These sandstone samples were

TABLE 1. Statistical Parameters of Paleomagnetic Data Based on the Analysis of As-Zijderveld Diagrams

	ΔT, °C	N	D, deg	I, deg	K	α_{95}	LAT, deg	LONG, deg	dp	dm
					Nicaragua					
Rivas Formation (Late Cretaceous)										
(1) 11.4°N, 274.2°E	220-400	18/2	352.8	32.4	72.3	4.1	80.7	226.8	2.6	4.6
			356.4	24.3	71.9	4.1	86.2	205.3	2.4	4.4
(2) 11.4°N, 274.2°E	220-400	11/2	1.6	21.7	262.4	2.8	88.4	10.4	1.6	3.0
			3.9	15.8	264.6	2.8	84.9	44.8	1.5	2.9
Brito Formation (Paleocene/Eocene)										
(3) 11.2°N, 274.2°E	220-400	20/11	176.3	−16.0	56.6	4.0	85.2	144.6	2.3	4.5
			178.9	−4.2	56.3	4.4	80.9	100.8	2.2	4.4
		6N	357.2	17.9	47.3	9.8	86.6	148.4	5.3	10.2
		14R	175.9	−15.1	58.1	5.3	84.7	143.6	2.8	5.4
					Costa Rica					
Rivas Formation (Campanian/Paleocene)										
(4) 10.9°N, 274.3°E	220-500	14/3	1.7	−8.6	63.6	5.0	74.8	91.5	2.6	5.1
			2.4	29.0	87.6	4.3	84.9	300.6	2.6	4.7
(5) 9.8°N, 275.0°E	220-500	16/0	357.6	−7.0	113.8	3.5	76.5	105.2	1.8	3.5
			357.4	15.1	145.2	3.1	86.7	147.2	1.6	3.2
Barra Honda Formation (Paleocene)										
(6) 10.2°N, 274.8°E	220-400	10/7	191.0	15.0	9.2	16.8	69.1	62.9	8.8	17.2
			193.7	4.0	9.2	16.8	74.1	35.0	8.4	16.8
		4N	23.1	3.4	9 9	30.8	65.5	23.6	15.4	30.8
		6R	187.3	−4.3	8.5	24.4	79.2	52.3	12.2	24.4
Brito Formation (Eocene)										
(7) 11.0°N, 274.3°E	220-500	27/11	1.0	−14.1	44 3	4.2	71.8	91.3	2.2	4.2
			0.3	22.8	65.3	3.5	89.1	291.7	2.0	3.7
		19N	3.4	−14.5	83.2	3.7	71.3	83.8	1.9	3.8
		8R	175.1	13.0	22.3	12.0	71.7	110.2	6.2	12.2
(8) 9.0°N, 276.6°E	450-540	9/14	184.7	3.3	140 9	4.4	78 4	72.8	2.2	4.4
			182.8	−20.0	140.0	4.4	87.0	340.4	2.4	4.6
	0-350	16/7	359.4	3.8	54.3	5.0	82.9	101.4	2.5	5.0

ΔT, temperature range used to calculate the directions; N, number of samples included/rejected in the analysis; N and R behind the numbers indicate normal or reversed directions; D, declination; I, inclination; K, precision parameter; α_{95}, radius of 95% circle of confidence; LAT, north latitude; LONG, east longitude; dp/dm, semi-axes of ellipse of confidence. For each site, the first entry is structurally corrected and the second entry is without tilt correction.

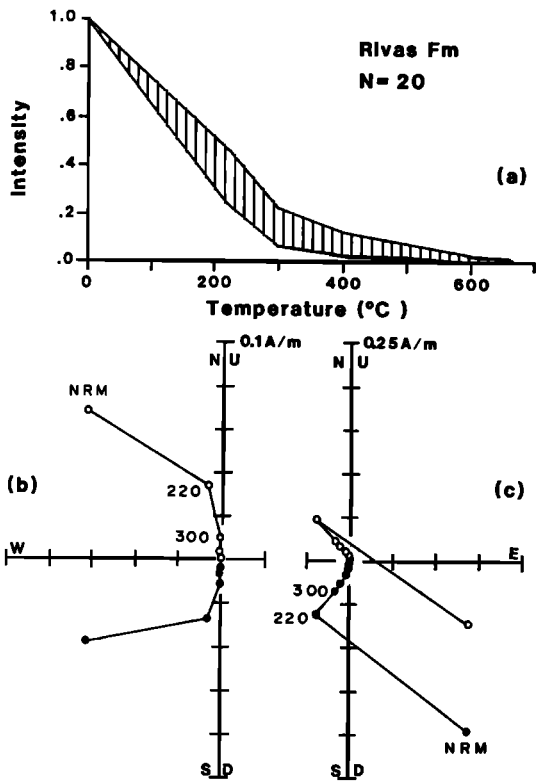


Fig. 2. Change in intensity of magnetization during thermal demagnetization (site 1). Shaded area delineates the total variation for all samples. The lower part of the figure shows two examples of As-Zijderveld diagrams. Open (solid) symbols are the projection on the horizontal (vertical) plane. The temperature range in Figure 2b is from NRM to 500°C, and in Figure 2c from NRM to 600°C. The higher-temperature steps have been deleted for clarity.

thermally demagnetized in eight steps to 665°C. Heating to 220°C reduced the intensity of magnetization to about 40% coupled with a sometimes pronounced change in direction (Figure 2). At this demagnetization step the directions clustered well and did not change significantly up to 550°, whence the intensities had decreased to less han 5% of their natural remanent magnetization (NRM) values. Upon exceeding the magnetite Curie point, the directions scattered.

The middle part of the overlying Brito Formation is well exposed near the harbor of San Juan del Sur (site 3). We collected 31 samples (sandstone/siltstone) over a 40-m stratigraphic interval. The broadly scattered NRM directions separate into a normal and reverse group after heating to 220°C and more clearly after 300°C (Figure 3). The reversed samples occur in three distinct stratigraphic zones. Eleven samples have "intermediate" directions, and all but two of these fall stratigraphically between the normal and reverse zones. The graph of declination versus stratigraphic position shows this most clearly (Figure 3). These 11 samples have been deleted from the statistics.

Costa Rica (Sites 4–8)

Our northernmost sampling sites in Costa Rica are located on the Santa Elena peninsula, 40–50 km south of the Nicaraguan sites. The Rivas Formation, exposed in the northern part of Santa Elena, is in fault contact with ultramafic rocks. Seventeen samples were collected at two outcrops 0.5 km apart (site 4). Upon demagnetization to 540°C, the samples changed their direction of magnetization by only a few degrees.

The second site in the Rivas Formation is located at the southeastern tip of the Nicoya peninsula (Figure 1, site 5). Sixteen sandstone samples from this site were demagnetized in eight steps up to 630°C. Between 220° and 500°C, a well-defined normal direction of magnetization is observed (Figure 4). At 500°C the intensities had decreased to 1–3% ($14–55 \times 10^{-4}$ A/m) of the NRM values. Upon demagnetization to higher temperatures, scattered, but clearly reversed directions prevail (Figure 4). The directions between the 220° and 500°C demagnetization steps were used for the analysis.

The reefal limestones from the Barra Honda Formation, collected near the Rio Tempisque ferry (site 6), are very weakly magnetized. The NRM intensities range from 14×10^{-6} to 14×10^{-4} A/m and after demagnetization to 400°C, the values range from 0.5 to 14×10^{-5} A/m. Ten of the 17 samples displayed consistent directions during demagnetization (220°, 300°, 350°, 400°C), as exemplified in Figure 5. Four samples are of normal, six of reverse polarity.

A road construction site on the Santa Elena peninsula offered the opportunity to collect unweathered samples over a 60-m stratigraphic interval (site 7). This section most likely exposes the basal part of the Brito Formation, although it is possible that it represents the upper part of the Rivas Formation. We collected 38 shale, mudstone, and fine-grained sandstone samples. Heating to 220°C removed a strong normal overprint (60–70% of NRM intensity), and upon further demagnetization the directions separated into two antipodal groups. The eight reversely magnetized samples occur at two stratigraphic horizons.

In southern Costa Rica the Brito Formation is exposed in road cuts of the Panamerican Highway [Henningson, 1966]. Although the highway was built only in 1962, the rocks are deeply weathered. East of Palmar, we collected 24 samples (site 8). The magnetization vectors erased between NRM and 300° or 350° yield a normal direction of magnetization for 16 samples. Nine of these samples exhibit a well-defined reversed direction in the temperature range from 450° to 540°, exactly antiparallel to the low-temperature component (Figure 6).

Discussion

The statistical parameters of the paleomagnetic analysis are listed in Table 1 and the pole positions are depicted in Figure 8. All pole positions fall within 21° of the rotational pole. It is important to ascertain that these directions reflect the primary magnetization and not a recent overprint, particularly because the results from 10 sampling sites had to be rejected due to poor data quality.

The structurally uncorrected pole positions (Figure 8a) cluster around the geographic pole, which may imply that all rocks have been remagnetized. While this possibility cannot be ruled out unequivocally, there are several lines of

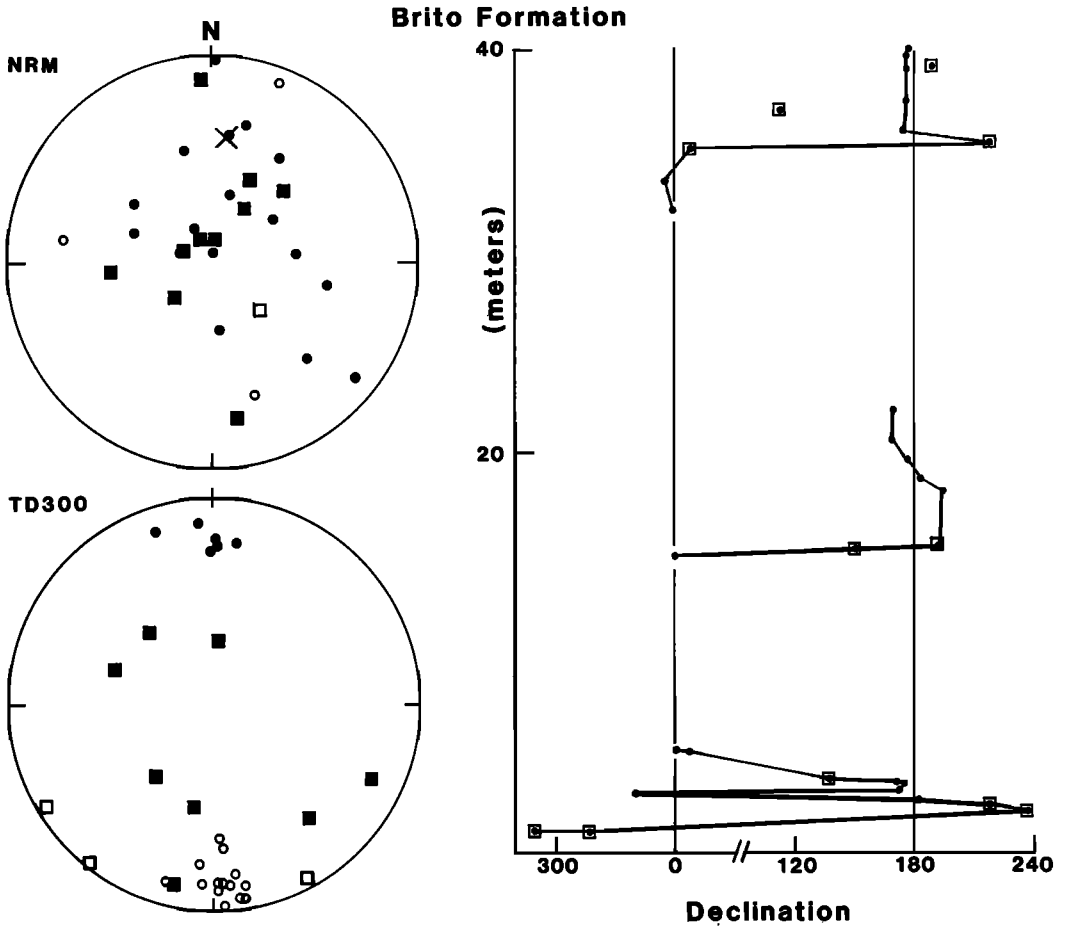


Fig. 3. Schmidt equal-area projection of directions of magnetization for the Brito Formation (site 3). NRM is plotted without structural correction and the 300° demagnetization step with the structural correction applied. Solid symbols are in the lower hemisphere, open symbols in the upper. Cross shows the present field direction. The squares indicate the samples which have been deleted from the statistics. The right part of the figure depicts the declinations after demagnetization to 300°C in stratigraphic order. Note that all but two of the deleted samples (squares) fall stratigraphically between zones of normal and reversed polarity.

evidence which argue for a primary magnetization. The pervasive weathering should impart a magnetization parallel to the present field direction rather than the theoretical dipole direction. However, the magnetic pole, calculated from the present values of declination and inclination for Costa Rica, is some 10° removed. Thus the magnetization is not the result of recent weathering. The strongest argument in favor of a primary magnetization is the occurrence of reversals. Not only are the reversely and normally magnetized samples truly antipodal for a given site (see Table 1), they also occur in distinct stratigraphic zones. At site 3 (Brito Formation in Nicaragua) we observed four normal and three reversed zones; at site 7 (Brito Formation in Costa Rica) there are three normal and two reversed zones; and at site 6 (Barra Honda Formation in Costa Rica) the samples extend over two normal and two reversed zones. The preservation of a magnetostratigraphic zonation almost certainly precludes remagnetization.

The thermal demagnetization experiments indicate the presence of goethite, magnetite, and hematite but clearly imply that the stable remanence is carried by magnetite. The presence of magnetite as the dominant magnetic mineral is confirmed by hysteresis measurements (Figure 7). All samples from the Rivas and Brito Formations exhibit the same behavior. The magnetization saturates in fields between 0.2 and 0.3 T, indicative of magnetite, and at higher applied fields (up to 2.0 T), all curves show a simple paramagnetic field dependence. For the three examples depicted in Figure 7, the concentrations of magnetite are 0.40, 0.21, and 1.82 wt %, values typical for all tested samples. Because it is very difficult to form magnetite authigenically, [e.g., Garrels and Christ, 1965], and because the Rivas and Brito Formations are volcanogenic sediments, the stable magnetization is almost certainly a detrital remanence, that is, it was acquired soon after deposition. Thus both the paleomagnetic as well as rock magnetic analyses indicate that these rocks carry their primary magnetization, and the following interpretation is based on this premise.

INTERPRETATION

The results from southern Nicaragua and from Costa Rica seem to form two distinct groups (Figure 8b). This is born out by applying the F test [Watson, 1956]. Excluding site 6, the results from all other sites have similar precision

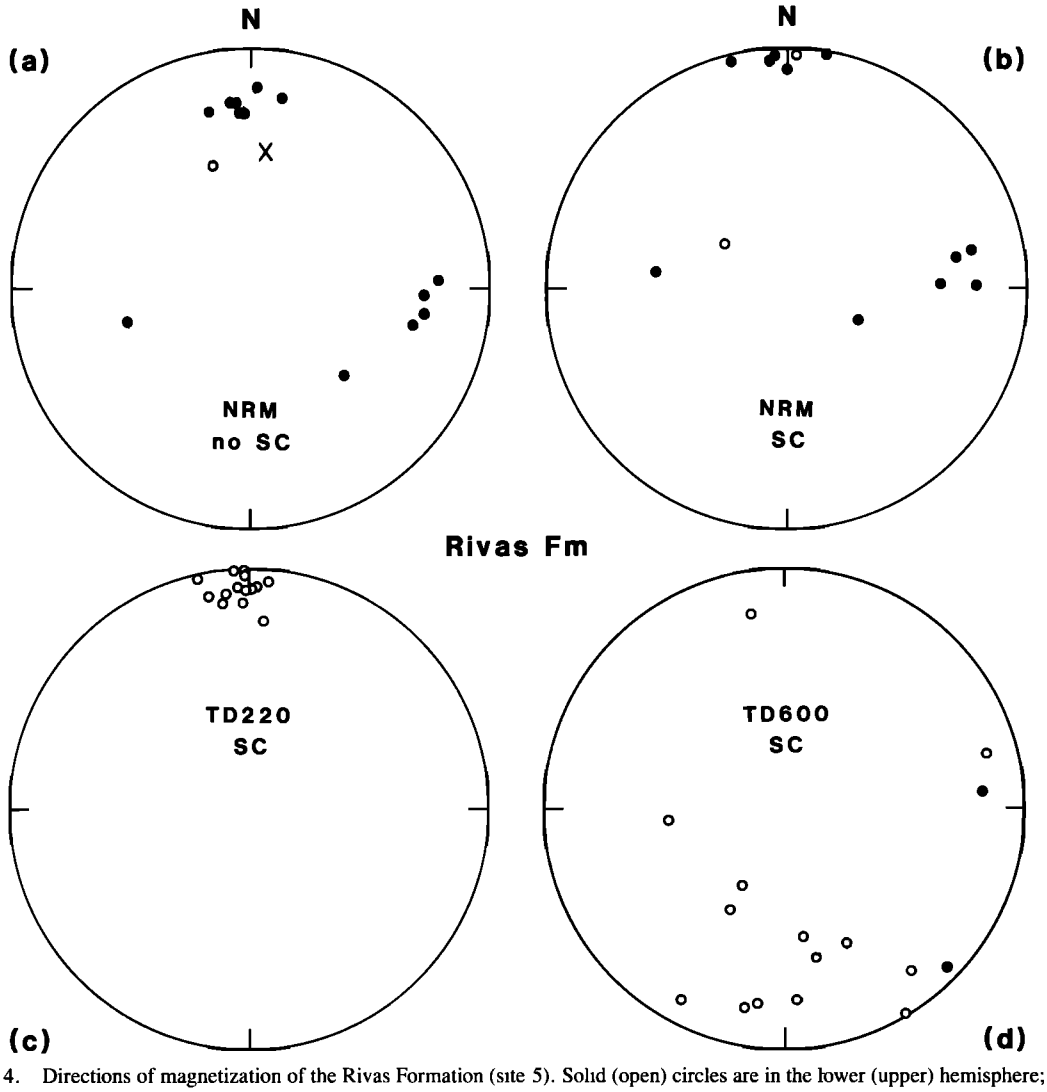


Fig. 4. Directions of magnetization of the Rivas Formation (site 5). Solid (open) circles are in the lower (upper) hemisphere; the cross represents the present field direction. (a) NRM without structural correction. (b) NRM with structural correction. (c) Directions after demagnetization to 220°C. (d) To 630°C. Figures 4c and 4d are structurally corrected. Schmidt equal-area projection.

parameters (K in Table 1) and can thus be tested whether or not they are samples from one population. At the 5% significance level the data indeed represent two distinct populations. This result is quite surprising because there is no major recognized fault between the Nicaraguan sites and the Santa Elena sites. There are, however, important stratigraphic differences. In Nicaragua the Rivas Formation ranges from Cenomanian through Maastrichtian and the Brito Formation from Middle Paleocene through Eocene, whereas in Costa Rica the Rivas Formation is restriced to the Campanian through Early (?) Paleocene and the Brito Formation to the Eocene. Thus in spite of great lithological similarities, they may not be the same formation as pointed out by Tax Improvement and Natural Resoures Inventory Project [1972].

All sampling sites are located in the Pacific coastal zone because most of the inland area is covered by young volcanics and is not sufficiently mapped. This leaves the question open whether the data apply only to this area or whether they are also valid for the interior region. The mean pole

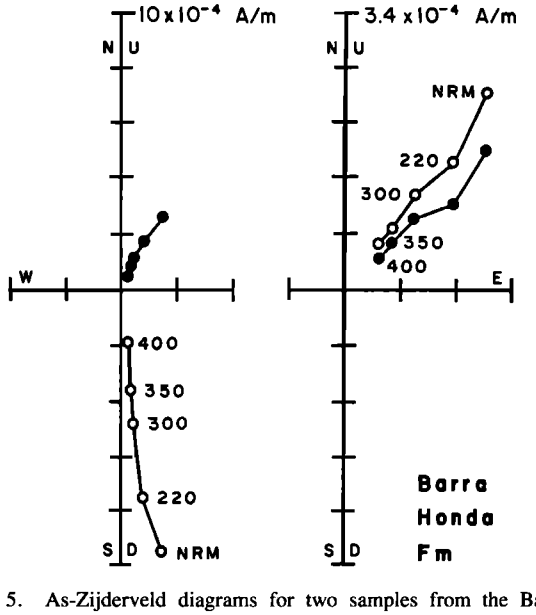


Fig. 5. As-Zijderveld diagrams for two samples from the Barra Honda Formation (site 6).

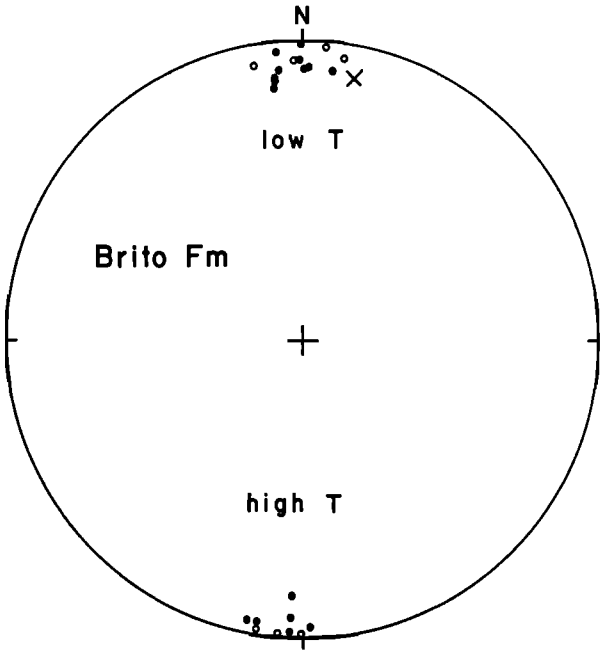


Fig. 6. Low- and high-temperature directions of magnetization of the Brito Formation (site 8) based on the analysis of As-Zijderveld diagrams. Low-temperature directions represent the vector which was erased between NRM and TD 300, and the high-temperature directions are the vector sum from 450° to 540°. Solid (open) symbols are in the lower (upper) hemisphere. Cross represents the present field direction rotated by the same amount as the bedding tilt. Schmidt equal-area projection.

position of the three Nicaraguan sites (86.8°N, 203.0°E, α_{95} = 9.8°) includes the geographic pole, which implies that the sampling area has been near or at its present latitude since the Late Cretaceous. The Costa Rican poles, on the other hand, call for a 14° northward translation in post-Eocene time (mean of all Costa Rican poles is 75.4°N, 73.6°E, α_{95} = 8.1°; mean of poles 4, 5, 7, 8 is 75.6°N, 90.0°E, α_{95} = 4.7°). On the basis of the paleomagnetic analysis of late Cretaceous basalt flows and intrusives of the Nicoya Complex, De Boer [1979] infers a similar northward translation.

What constraints do these data place on the location of the boundary between northern (nuclear) and southern Central America, and what can be inferred about early plate motions? All our data from Costa Rica agree closely. It is therefore unlikely that a major boundary lies in this region. The results from the southern Nicaraguan sites, on the other hand, are significantly different from the Costa Rican data and indicate north-south convergence of the two areas. This is supported by other geological observations. For example, the region near the Santa Elena peninsula is "something of a foreign body in the structure of Costa Rica" [Weyl, 1980, p. 143]. The structural grain of Nicaragua and Costa Rica trends mainly northwest-southeast [e.g., Case and Holcombe, 1980], whereas the Santa Elena area is characterized by east-west trending faults and folds (mapa geologico generalizado de la provincia de Guanacaste y zonas adyacentes [Dengo, 1962; Tax Improvement and Natural Resources Inventory Project, 1972, Plate IV-3A; Weyl, 1980]). De Boer [1979] interprets the east-west faults on Santa Elena as thrust faults arising from north-south compression and also recognizes a left-lateral strike-slip component.

I do not believe, however, that these faults define the boundary between northern and southern Central America, as proposed by de Boer [1979] and as used by Pindell and Dewey [1982]. One reason is that the pole positions from southern Nicaragua as well as from Costa Rica are very different from the pole positions from Honduras. The Late Cretaceous data from Honduras [Gose and Swartz, 1977], require a ~70° rotation between northern and southern Central America, and earliest Paleocene data from Honduras (W. A. Gose, manuscript in preparation, 1983) call for a ~50° rotation. This suggests that the major boundary between these two tectonic provinces must lie to the north and east of our Nicaraguan sampling sites. It seems likely that the Nicaraguan Depression delineates this boundary because it separates the Cretaceous-Tertiary sedimentary basin of the Pacific coast from Tertiary volcanic plateau of the interior highland. As has been pointed out by many authors [e.g., Dengo and Bohnenberger, 1969; Weyl, 1980], it may not be possible to define the boundary closely due to the extensive volcanic cover.

As mentioned earlier, all sampling sites are located in the Pacific coastal zone, and it seems probable that the paleomagnetic data pertain only to this area. The Late Cretaceous and Early Tertiary sediments which overlie the Nicoya Complex were deposited in a subduction related environment, and this intraoceanic subduction began in the Latest Cretaceous [Schmidt-Effing, 1979; Kuijpers, 1980; Lundberg, 1982]. According to the paleomagnetic data, this subduction zone was near the equator from Campanian through Eocene time. The subsequent northward movement cannot have been initiated by the development of the Cocos-Nazca spreading center because it became active only 25 m.y. ago [Hey, 1977], and the crust, which was located near the equator at that time, is only now being subducted along the Middle America Trench [von Huene et al., 1980; Gose, 1982]. It seems likely that the extensive late Miocene plutonic activity [Weyl, 1980] defines the time of collision be-

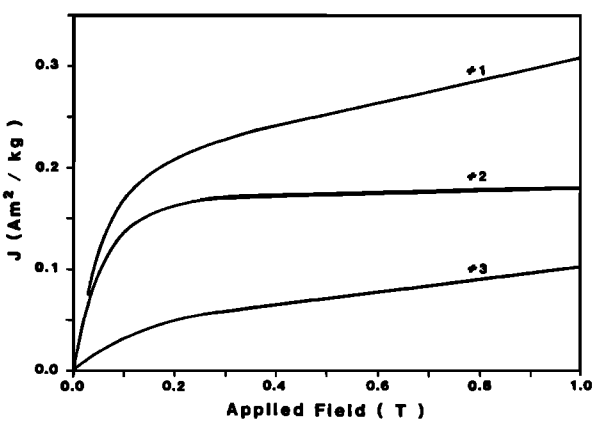


Fig. 7. Magnetization as a function of applied field for two samples from the Brito Formation (1, 3) and one sample from the Rivas Formation (2). The magnetization of sample 2 is plotted at one tenth the actual value.

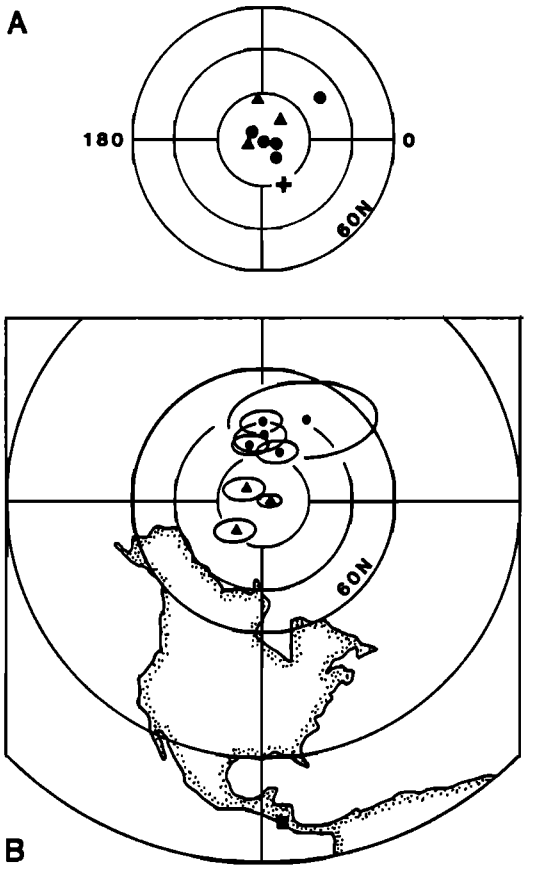


Fig. 8. Pole positions from southern Central America (a) without and (b) with structural corrections. Triangles represent the results from Nicaragua, circles the results from Costa Rica, and the cross marks the pole position based on the present field direction for the sampling area (square).

tween the Nicoya Complex (sensu largo) and the Costa Rican island arc.

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